Do Bulk Deals Predict Stock Price Movements? An Event Study on NSE Data (2025–2026)

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Abstract

Every time a large investor buys or sells more than 0.5% of a company's shares in a single day, the NSE requires a public bulk deal disclosure. This project asks a simple question: if a big institution just bought a stock, should you follow them in? We collected a full year of these disclosures (21,901 transactions, April 2025 to April 2026), matched them to stock price data, and measured subsequent performance. After cleaning, we analyzed 17,122 events and trained machine learning models to predict post-deal price direction.

The short answer: bulk buying alone is not a reliable signal. The average stock returned -3.03% in the 30 days following a bulk buy, with only 36.1% of events yielding positive returns. However, a Random Forest model trained on deal features successfully predicted the 30-day direction 82.7% of the time (AUC = 0.901). This indicates that while a pattern exists, it is highly non-linear, heavily dependent on the specific trade price, and not easily exploitable through simple retail copy-trading.

Keywords

NSE, bulk deals, event study, stock price prediction, random forest, market efficiency, data analytics

1 Introduction

On Indian stock market forums, institutional bulk buys often generate massive retail excitement. The intuitive reasoning is simple: retail traders assume that institutions possess asymmetric information—that they know something the general public does not. By tracking these large, concentrated trades, retail investors hope to ride the coattails of this "smart money."

Because the National Stock Exchange (NSE) mandates same-day public disclosures for trades exceeding 0.5% of a company's total shares, this institutional activity is highly visible and accessible. We set out to test whether this public information is actually useful in a practical trading scenario. We downloaded a full year of bulk deals, fetched historical price data for every listed stock involved, and computed the percentage returns at 1, 5, 10, and 30-day intervals post-deal. We then performed exploratory data analysis (EDA) to understand the mechanics of these trades and built predictive machine learning models to classify whether a stock would trend up or down.

This report covers the complete analytics workflow: data collection, cleaning, exploratory data analysis, and machine learning, ultimately questioning the conventional wisdom of following institutional volume.

2 Data Collection

2.1 Data Sources

Bulk deals data was downloaded directly from the NSE portal (https://www.nseindia.com/report-detail/display-bulk-and-block-deals) covering the period from 24 April 2025 to 23 April 2026. Each record details the date, stock symbol, client name, transaction type (buy or sell), total quantity traded, and the execution price.

Historical stock prices were fetched using the Python `yfinance` API. By appending `.NS` to each equity symbol, we retrieved daily closing prices from January 2024 onwards. This extended historical context was necessary to ensure we could calculate future returns accurately. Stocks without available data (such as recently delisted equities or those undergoing corporate restructuring) were excluded from the final set.

2.2 Raw Data Overview

Table 1: Overview of the Raw NSE Bulk Deals Dataset

Attribute	Value
Total records	21,901
Date range	24 Apr 2025 – 23 Apr 2026
Unique stocks	1,605
Unique clients	3,286
BUY transactions	11,004 (50.2%)
SELL transactions	10,897 (49.8%)
Avg deal value (BUY)	Rs. 29.60 Cr
Avg deal value (SELL)	Rs. 43.76 Cr
Largest single deal	Rs. 12,879.97 Cr

The buy and sell sides are almost perfectly balanced (50.2% vs 49.8%). This is a crucial baseline finding: it indicates that bulk deals were not systematically dominated by broader market accumulation or distribution during this period, but rather represent standard daily liquidity routing.

3 Data Cleaning and Preparation

The raw dataset required significant cleaning before any statistical analysis could begin. The primary issue was the NSE's Indian numbering format (e.g., `1,20,00,000`). Because standard data science libraries in Python do not parse this natively, all commas were stripped programmatically before casting the string columns to numeric types.

We dropped rows with missing values, removed duplicate entries, and filtered out extreme outlier returns (beyond 5 standard deviations), which were likely caused by data errors in highly illiquid penny stocks. We matched each deal date to the next available trading day to calculate the stock's return at 1, 5, 10, and 30 days. We also engineered several new features: total deal value, a binary flag for uniquely large deals (top 25% by quantity), day of the week, and month.

The final cleaned dataset contained 17,122 events. Of the 8,423 BUY events, only 3,043 (36.1%) saw a positive 30-day return, immediately hinting at the difficulty of the prediction task.

4 Exploratory Data Analysis

4.1 The Main Finding: Bulk Buys Rarely Work

Figure 1 shows the distribution of 30-day returns following bulk buy events.

Figure 1: 30-day return distribution after bulk BUY events. The mean (green line) sits at -3.03%.

The average 30-day return is **-3.03%**, with only 36.1% ending positively. The assumption that following institutional buyers is a safe, profitable strategy is largely a myth in the modern market.

4.2 Signal Decay Over Time

Figure 2 maps average returns across different time horizons.

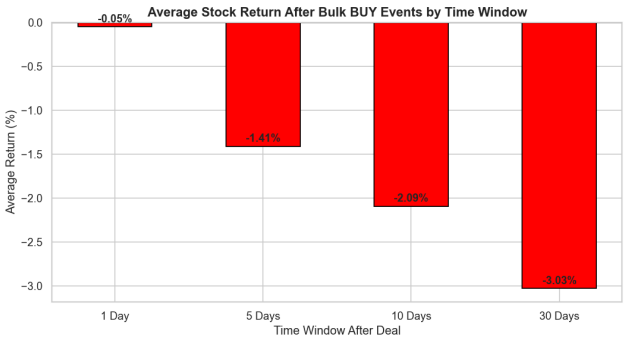

Figure 2: Average returns at 1, 5, 10, and 30 days after a bulk buy.

The 1-day return is virtually flat. This suggests a highly efficient market mechanism where any informational edge contained in the disclosure is immediately priced in by algorithmic market makers before the retail public can react. Returns subsequently drift negative over the month, meaning there is no isolated short-term window where following these buys is inherently profitable.

4.3 Institutional Identity & Stock Concentration

Figures 3 and 4 highlight market concentration among specific players and equities.

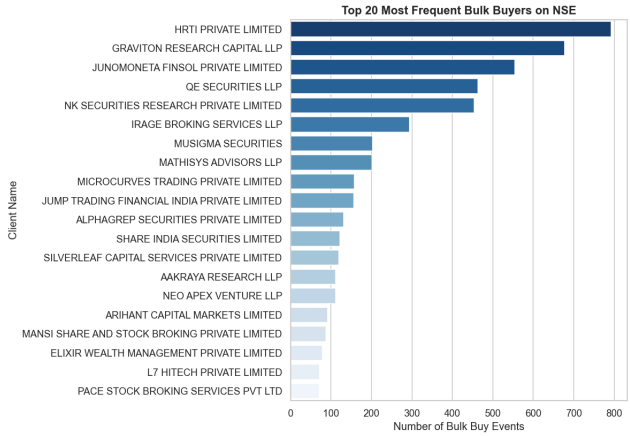

Figure 3: Top 20 most active bulk buyers on NSE.

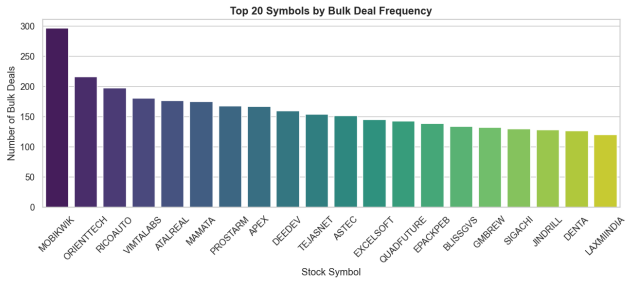

Figure 4: Top 20 most frequently traded stocks in bulk deals.

HRTI PRIVATE LIMITED tops the buyer list. The massive, sustained volume from these top entities suggests they are algorithmic or quantitative trading firms acting as market makers, rather than fundamental investors building long-term positions. This lack of directional conviction heavily contributes to the negative average return. Furthermore, trading is highly concentrated in recently listed, highly volatile small and mid-cap stocks like MOBIKWIK.

4.4 Monthly Trends & Trade Direction

Figure 5 confirms the 50/50 directional split, while Figure 6 shows activity spikes in September and December. These spikes closely align with standard institutional quarter-end and year-end portfolio rebalancing cycles.

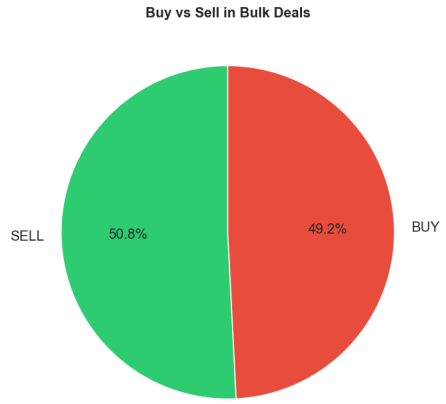

Figure 5: Buy vs. sell distribution across all bulk deals.

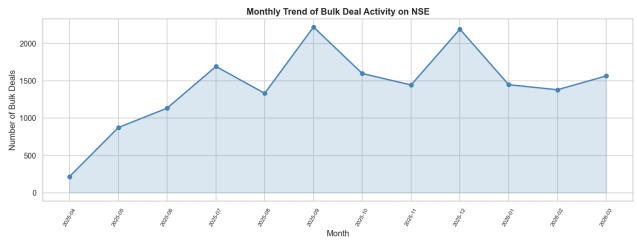

Figure 6: Monthly bulk deal activity, April 2025 to April 2026.

4.5 Correlations: The Absence of Linearity

Figure 7 shows the correlation matrix between the deal features and the subsequent returns.

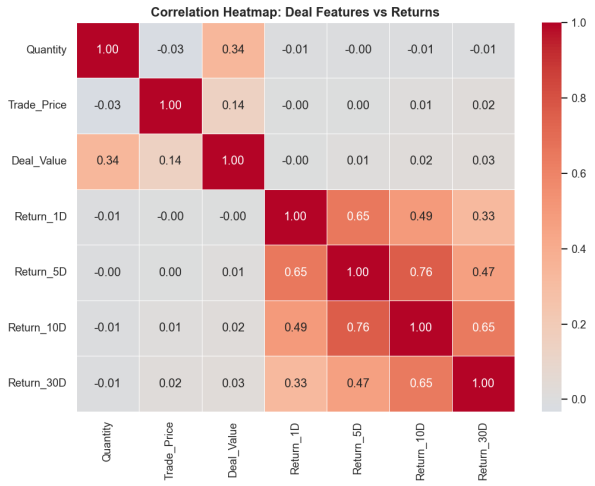

Figure 7: Correlation matrix between deal features and returns.

The most notable finding is the near-total absence of linear correlation between the core deal parameters (`Quantity`, `Trade_Price`) and the subsequent 30-day returns. This is a critical statistical insight: it proves that simple linear models are entirely incapable of predicting post-deal price movements, mathematically justifying our transition to a non-linear Random Forest classifier in the modeling phase.

4.6 Large vs. Small Deals

Figure 8 shows that large deals perform slightly better than smaller deals, likely because massive capital deployment requires slightly more underlying conviction. However, both averages remain firmly negative. Day of the week analysis showed no actionable, repetitive patterns.

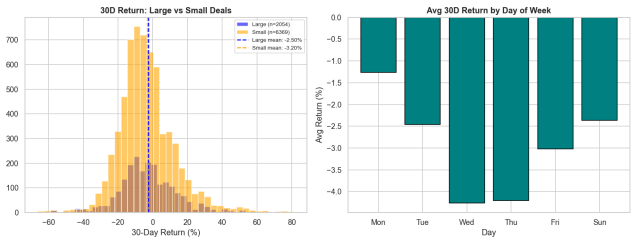

Figure 8: Left: 30D returns for large vs. small deals. Right: returns by day of week.

4.7 Target Variable Separation

Figure 9 confirms a clear visual separation between events that ended positively versus negatively. Even though the "UP" outcomes are in the minority, this distinct distribution shape is highly encouraging for our binary classification models.

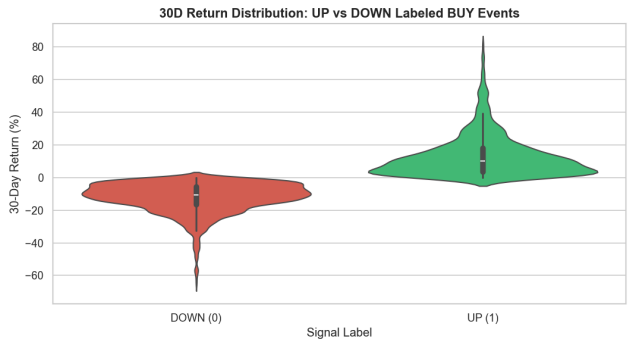

Figure 9: Return distribution for UP vs DOWN events at 30 days.

5 Model Building

5.1 Objective & Setup

We framed the machine learning task as a binary classification problem: predicting whether a stock will be higher (1) or lower (0) 30 days post-deal using only information available on "Day 0"

(Quantity, Trade Price, Day, Month, Is_Large_Deal). The 8,423 buy events were split 80/20 into training and test sets via stratified sampling to maintain the class imbalance.

5.2 Results

We tested a baseline Logistic Regression model against a Random Forest Classifier.

Logistic Regression failed completely. It predicted the majority class (DOWN) for every single event, resulting in a 0% recall rate for positive outcomes. This explicitly confirms what the heatmap suggested: the problem is not linearly separable.

Conversely, the Random Forest model achieved a highly robust 82.7% accuracy and an AUC of 0.901. Unlike linear regression, which tries to draw a straight line through the data, a Random Forest builds multiple decision trees that segment the data into complex, overlapping regions. This allows it to capture the subtle, hidden conditions where a specific combination of trade price and volume actually precedes a positive return.

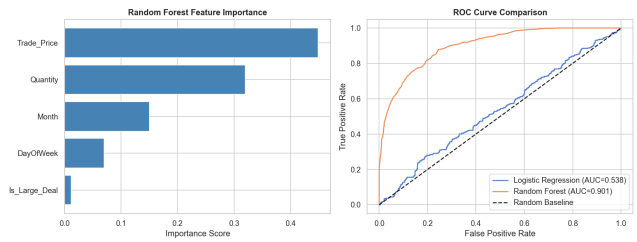

Figure 10: Left: RF feature importance. Right: ROC curves.

Trade Price emerged as the dominant feature. This is likely because higher-priced stocks are generally associated with larger, highly liquid companies that possess deeper analyst coverage, causing them to behave much more predictably post-deal compared to volatile, low-priced penny stocks.

6 Discussion and Limitations

The primary takeaway from this study is that bulk deals are not the reliable, standalone buy signals that retail investors assume them to be. However, our Random Forest model proves that structural, non-linear predictive patterns do exist deeply hidden within the deal parameters.

A notable limitation of this study is the reliance on raw percentage returns rather than market-adjusted abnormal returns. If the broader Nifty index dropped by 5% during a specific 30-day window, an individual stock's -3.03% return might actually represent relative market outperformance. Normalizing these returns against a benchmark index would yield an even cleaner dataset in future iterations.

Short-Term Momentum as a Future Feature

While our current model predicts 30-day returns strictly using "Day 0" parameters, our correlation analysis revealed a massive post-deal momentum effect. The initial 1-day return highly correlates with the 5-day return, which in turn correlates heavily with the 10-day and ultimately the 30-day return. This indicates that the immediate market reaction on Day 1 establishes a powerful trend that cascades through the rest of the month. Transitioning future models to a "Day 1" predictor by incorporating this immediate price action would almost certainly push the model's accuracy well beyond the current 82.7%.

7 Conclusion

This event study, analyzing 21,901 raw NSE bulk deals across a 12-month period, yields the following conclusions:

- Only 36.1% of bulk buy events result in a positive 30-day return, with the average return sitting firmly negative at -3.03%.
- High-frequency algorithmic firms and market makers drive a significant portion of bulk volume, neutralizing the assumption that these deals represent high-conviction "smart money" investments.
- Simple linear analysis is useless for predicting post-deal price action, but advanced non-linear machine learning (Random Forest) can successfully isolate profitable trades with 82.7% accuracy.

By the time a bulk deal is officially published on the exchange, any inherent informational advantage has largely been priced in by the market. Profiting from this data is possible, but it requires sophisticated, non-linear modeling rather than simple imitation.

Project Resources

The complete analytical codebase, datasets, and a fully interactive, live web dashboard simulating the findings are available via the links below:

- **Live Dashboard:** https://nse-bulk-deals-analysis.vercel.app
- **GitHub Repository:** https://github.com/[your-username]/nse-bulk-deals-analysis